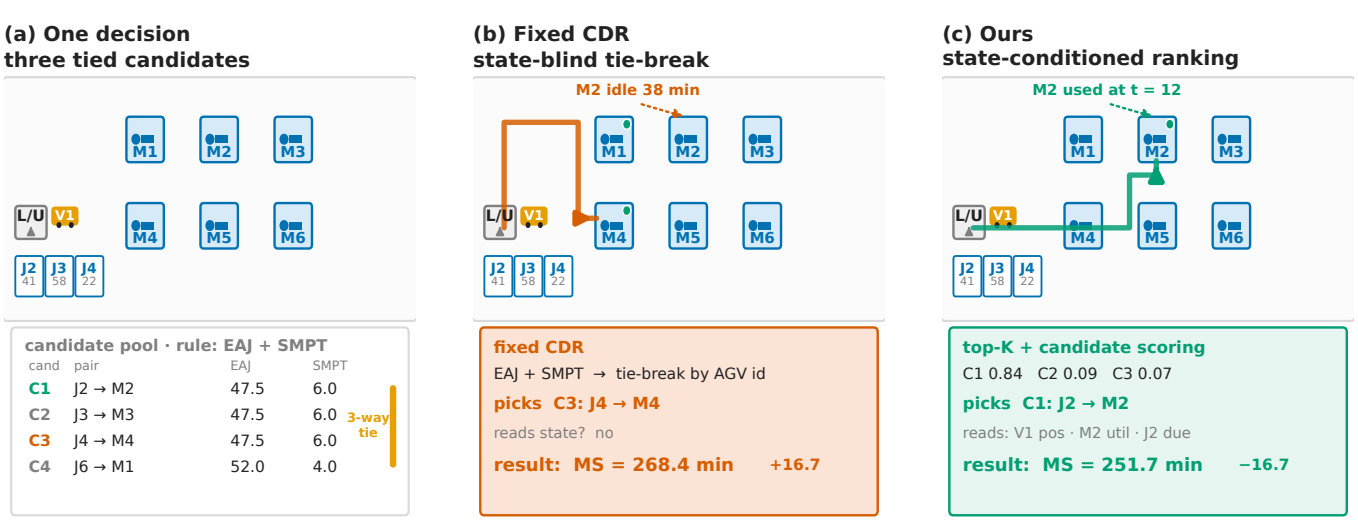

(a) One decision
three tied candidates
M1
M2
M3
L/U
V1
M4
M5
M6
J2
41
J3
58
J4
22
candidate pool · rule: EAJ + SMPT
cand
pair
EAJ
SMPT
C1
J2 → M2
47.5
6.0
C2
J3 → M3
47.5
6.0
C3
J4 → M4
47.5
6.0
C4
J6 → M1
52.0
4.0
3-way tie
(b) Fixed CDR
state-blind tie-break
M2 idle 38 min
fixed CDR
EAJ + SMPT → tie-break by AGV id
picks C3: J4 → M4
reads state? no
result: MS = 268.4 min
+16.7
(c) Ours
state-conditioned ranking
M2 used at t = 12
top-K + candidate scoring
C1 0.84 C2 0.09 C3 0.07
picks C1: J2 → M2
reads: V1 pos · M2 util · J2 due
result: MS = 251.7 min
−16.7